# FD004 Cross Subset Transfer Study

Train on the six-condition, two-fault FD004 subset and evaluate locked transfer to FD001, FD002, and FD003

Prepared 20 October 2026 | CPU-bounded local execution

## Study design

FD004 is the source because it contains six operating conditions and two degradation modes. The model selection phase uses FD004 training engines only: three engine-grouped folds and six predeclared pseudo-censoring horizons (10, 25, 50, 75, 100, and 125 cycles). The condition transformer is fitted only on FD004 settings and sensors. Target-subset labels are not used for clustering, normalization, features, or training.

## FD004 validation selection

| Candidate | RMSE | MAE | R2 | PHM08 mean |
|---|---|---|---|---|
| fd004_hgb_cap100_condition_normalized | 15.323 | 10.739 | 0.856 | 3.460 |
| fd004_hgb_cap125_condition_normalized | 13.773 | 9.508 | 0.884 | 4.392 |
| fd004_hgb_cap125_raw_sensor_control | 19.278 | 14.483 | 0.772 | 9.510 |

The FD004-selected source model is HGB with a 100-cycle target cap and FD004-fitted condition normalization. Compared with the raw-sensor cap-125 control, condition normalization reduces validation RMSE from 19.278 to 15.323 and PHM08 mean from 9.510 to 3.460.

## Locked cross-subset endpoint results

| Target | Engines | RMSE | MAE | R2 | PHM08 mean | Bias |
|---|---|---|---|---|---|---|
| FD001 | 100 | 17.619 | 12.866 | 0.820 | 3.824 | -8.066 |
| FD002 | 259 | 31.074 | 19.194 | 0.666 | 62.550 | -14.266 |
| FD003 | 100 | 18.162 | 13.019 | 0.808 | 4.528 | -5.613 |

## Interpretation

FD001 and FD003 are single-condition target domains and transfer with moderate error. FD002 is the hardest transfer target: the source model underpredicts more often and the asymmetric PHM08 penalty is high. This is evidence of domain shift, not a failure of endpoint alignment or a reason to tune on FD002 labels.

## Recommended use

Present FD004 as the final multi-condition source-model demonstration. Present the FD001–FD003 figures as honest zero-target-training transfer tests. For a production candidate in a particular target domain, validate or fine-tune only with that domain's labelled training engines, retain an engine-grouped holdout, and calibrate RUL thresholds with asset owners.

## Reproducibility

Runner: run_fd004_transfer_study.py. Detailed source/target audit, per-target endpoint predictions, split files, selection table, and serialized source model are stored in reports/fd004_transfer and artifacts/FD004/transfer.